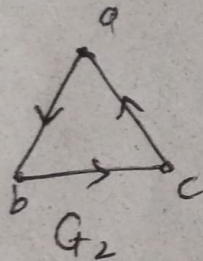
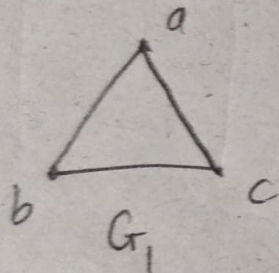


Graph

Theory

$$G(V, E)$$



$$G_1: V = \{a, b, c\}$$

$$E = \{(a, b), (b, a), (a, c), (c, a), (b, c), (c, b)\}$$

$$G_2: V = \{a, b, c\}$$

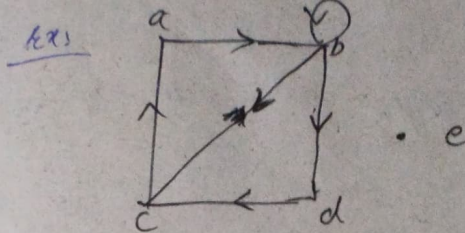
$$E = \{(a, b), (b, c), (c, a)\}$$

Graph:

Let V be a finite, non-empty set and E is a subset of $V \times V$ [$E \subseteq V \times V$], the ordered pair $G(V, E)$ is called as a graph.

Incidence:

Incidence in a directed graph:- For any edge say (b, c) we say that edge is incident with the vertices b and c and b is said to be adjacent to c whereas c is adjacent from b .



Loop

An edge where initial vertex and terminal vertex are same is called as loop or self-loop.

Isolated vertex

In a graph G , the vertex v that has no incident edges is called as isolated vertex.

"Degree of an isolated vertex is zero 0."

Degree of a Vertex

It is defined as The degree of a vertex v of the graph is denoted by $\deg_G(v)$ is the number of edges incident with vertex v , with self loop counted twice.

In degree :- No. of edges to which v is the terminal vertex.

Out degree :- No. of edges to which v is the initial vertex.

Ex 1

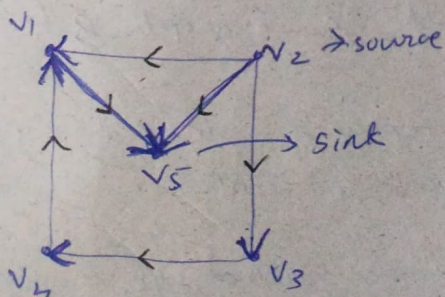
$$\deg_G(a) = 1_{in} + 1_{out} = 2$$

$$\deg_G(b) = 3_{in} + 2_{out} = 5$$

Sink and Source :-

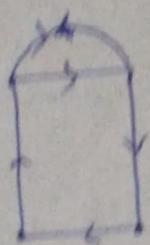
* A non-isolated vertex which has no incoming ^{edge} vertex is called as "source".

* A non-isolated vertex which has no outgoing edge is called as "sink".

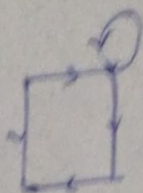


Simple graph:

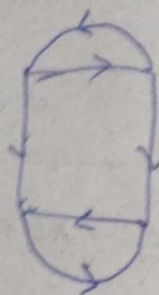
A graph with no multiple edges and self loop is called as a simple graph.



x : multiple edges



x : self loop



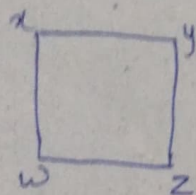
v : Simple graph

Null graph:

A graph without any edges or degree of all the vertices in a graph is zero then we call the graph as null graph.

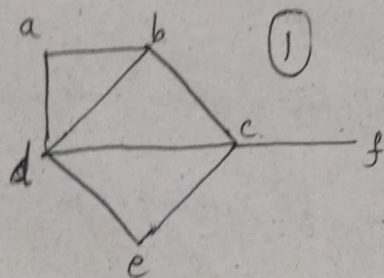
Common terms:

1) Walk / chain: Let x, y be the vertices in a undirected graph G . An x - y walk in G is a loop-free finite alternative sequence of vertices and edges in G starting at vertex ' x ' and ending at vertex ' y '.



walk x to z :

$x-y-z$ or $x-w-z$



Ex: walk of length 3

$a-b-c-f$

walk of length 4

$a-b-d-c-f$

2) Trail: Consider any x - y walk in a graph G where

i) Trail: If no edge is repeated in a x - y walk then it is called a trail.

ii) Circuit: A closed x - x trail is called as a circuit

iii) of no Path: If no vertex of $x-y$ walk is repeated then we call the walk as a path.

iv) cycle: The closed $x-x$ path is called a cycle.

Examples See figure ①

* b-f walk: 1) $b \rightarrow c \rightarrow d \rightarrow e \rightarrow c \rightarrow f$
trail but not path ($\because c$ repeats)

2) $b \rightarrow c \rightarrow f$ is trail ~~and~~ and path

* f-a walk:

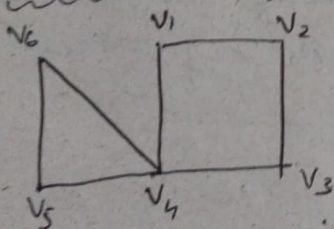
$f \rightarrow c \rightarrow e \rightarrow d \rightarrow a$ [trail & path]

* a-a walk:

$a \rightarrow b \rightarrow d \rightarrow c \rightarrow e \rightarrow d \rightarrow a$ } $a \rightarrow b \rightarrow d \rightarrow a$
Circuit but not a cycle. } circuit & cycle.

~~also~~

Indicate the following for the graph shown below:



1) $v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4$
~~Trail~~ Trail & path

2) $v_4 \rightarrow v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4 \rightarrow v_5$
Trail & not a path

3) $v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4 \rightarrow v_5$
Trail and path

4) $v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4 \rightarrow v_1$
circuit and cycle

5) $v_6 \rightarrow v_5 \rightarrow v_4 \rightarrow v_3 \rightarrow v_2 \rightarrow v_1 \rightarrow v_4 \rightarrow v_6$
circuit & not a cycle

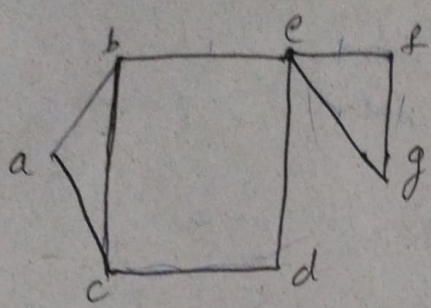
Determine a walk from:-

a) b-d that is not a trail
 $b \rightarrow c \rightarrow a \rightarrow b \rightarrow c \rightarrow d$

b) b-d trail which is not path
 $b \rightarrow e \rightarrow f \rightarrow g \rightarrow e \rightarrow d$

c) b-d path
 $b \rightarrow c \rightarrow d$

- d) $b-b$ closed walk that is not circuit
 $b \rightarrow a \rightarrow c \rightarrow a \rightarrow b$ or $b \rightarrow a \rightarrow c \rightarrow b \rightarrow e \rightarrow d \rightarrow c \rightarrow b$
 or $b \rightarrow e \rightarrow f \rightarrow g \rightarrow c \rightarrow b$
- e) $b-b$ circuit not a cycle
 $b \rightarrow e \rightarrow f \rightarrow g \rightarrow c \rightarrow d \rightarrow c \rightarrow b$
- f) $b-b$ cycle. $b \rightarrow a \rightarrow c \rightarrow b$



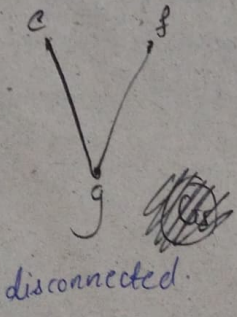
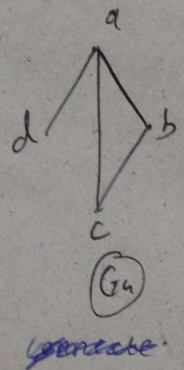
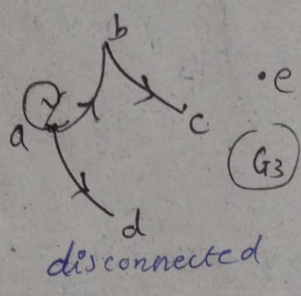
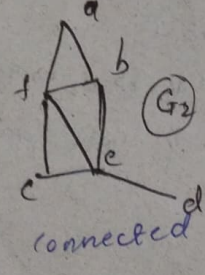
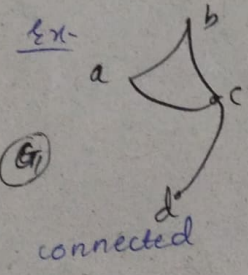
g) How many paths are there from b to f ?

- 6) 1) $b \rightarrow e \rightarrow f$ 2) $b \rightarrow e \rightarrow g \rightarrow f$
 3) $b \rightarrow c \rightarrow d \rightarrow e \rightarrow f$ 4) $b \rightarrow c \rightarrow d \rightarrow e \rightarrow g \rightarrow f$
 5) $b \rightarrow a \rightarrow c \rightarrow d \rightarrow e \rightarrow f$ 6) $b \rightarrow a \rightarrow c \rightarrow d \rightarrow e \rightarrow g \rightarrow f$

Connected graphs:

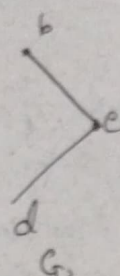
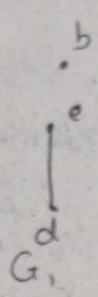
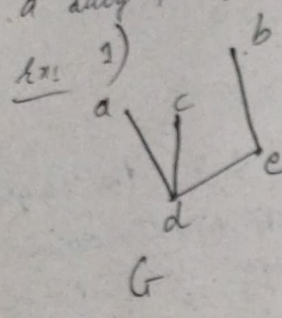
A graph G is said to be connected if there is atleast one path between every pair of vertices in G otherwise it is disconnected

ex-



Sub Graph:-

If G is a graph (directed or undirected) the G_1 which has vertices set V_1 and edge set E_1 is called as a subgraph of G if $\phi \neq V_1 \subseteq V$ and $E_1 \subseteq E$



$V = \{a, b, c, d, e\}$

$E = \{(a,d), (c,d), (d,e), (b,e), (c,e)\}$

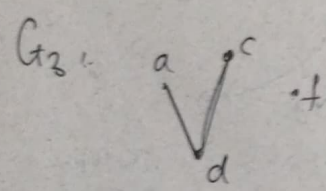
$V_1 = \{b, d, e\}$

$E_1 = \{(e,d), (d,e)\}$

Subgraph

$V_2 = \{b, d, e\}$

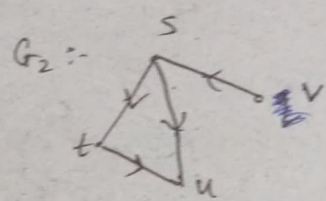
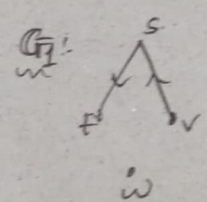
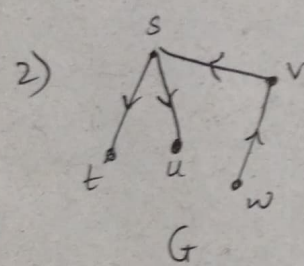
$E_2 = \{(b,e), (c,e), (c,d), (b,c), (c,b)\}$



$V_3 = \{a, c, d, e\} \not\subseteq V$

B_3

Not a subgraph



$V_1 = \{s, t, v, w\}$

$E_1 = \{(s,t), (v,w)\}$

Sub graph

$V_2 = \{s, t, v, u\}$

$E_2 = \{(s,t), (s,u), (t,u), (v,w)\}$

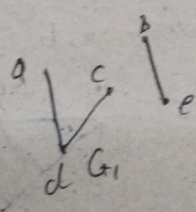
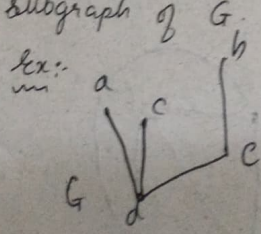
Not a subgraph

$V = \{s, t, u, v, w\}$

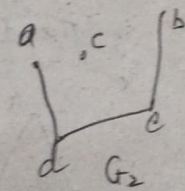
$E = \{(s,t), (s,u), (v,w), (w,u)\}$

Spanning subgraph

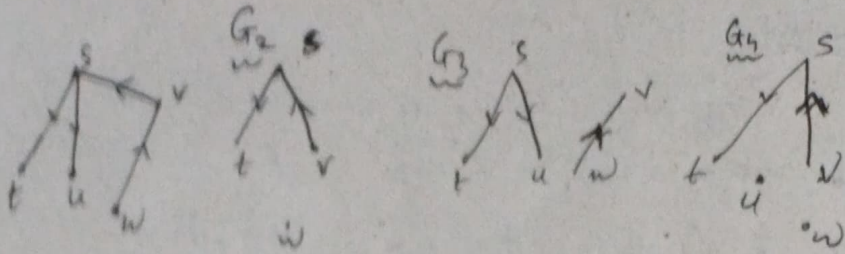
Given a graph G , let G_1 be the subgraph of G if vertex set $V_1 = V$ then G_1 is called as spanning subgraph of G



Spanning subgraph



Spanning subgraph

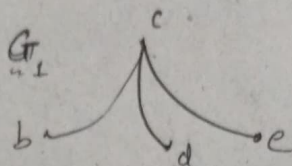
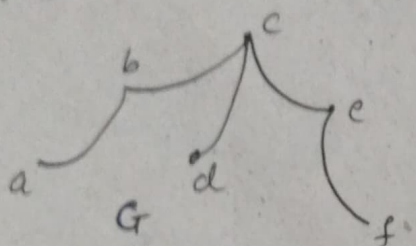


Induced Graph

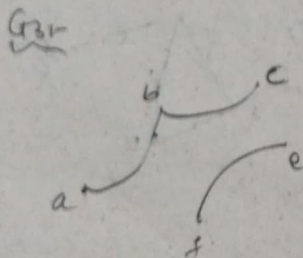
Let G be the graph of $\mathcal{U}$ where $\mathcal{U} \subseteq V$, the subgraph of G induced by $\mathcal{U}$ is a subgraph whose vertex set is $\mathcal{U}$ and which contains all the edges from G either of the form (x, y) or $\{x, y\}$ where $x, y \in \mathcal{U}$.

undirected directed

This subgraph is denoted by $\langle \mathcal{U} \rangle$ which

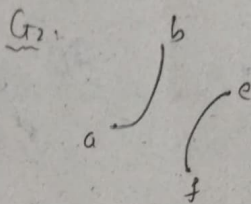


$\mathcal{U}_1 = \{b, c, d, e\}$ $\langle \mathcal{U}_1 \rangle$ induced



$\mathcal{U}_2 = \{a, b, c, e, f\}$

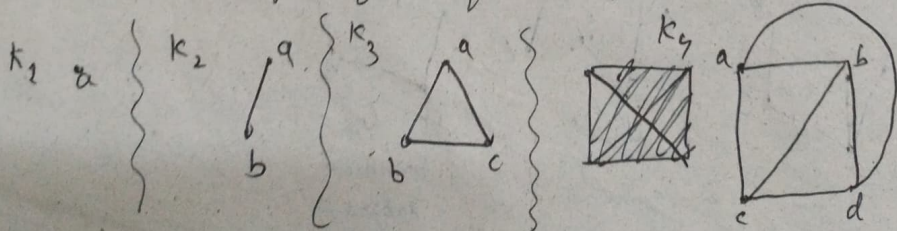
not induced

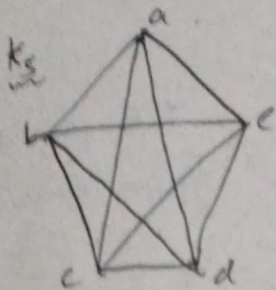


$\mathcal{U}_3 = \{a, b, e, f\}$ $\langle \mathcal{U}_3 \rangle$ induced

Complete graph

Let V be the set of n vertices. the complete graph on vertex set V denoted as K_n is a loop free undirected graph where $\forall (a, b) \in V$, and $a \neq b$, there is a edge of the form a, b

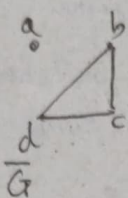
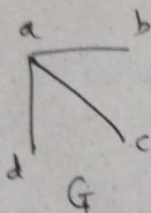




Complement of a graph:

Let G be the loop free undirected graph on n vertices, the complement of G which is denoted by $\bar{G}$ is a subgraph of K_n which consists of n vertices in G and all the edges ~~are~~ that are not in G .

Ex:



Isomorphic Graphs

Two graphs G and G' are isomorphic if there is one to one correspondence between their vertices and between their edges, such that ~~ed~~ incidence relation is preserved. In other words suppose that the edge is incident on v_1 and in graph G . Then the corresponding edges E' in G' must be incident on the vertices v_1' and v_2' respectively.

Pendant vertex:-

A vertex with degree 1 is called as pendant vertex or end vertex and the edge incident on it is called as pendant edge.

Isomorphic Graphs

Let G_1 and G_2 be two undirected graphs. A function which is mapped from V_1 to V_2 is called graph isomorphism if:-

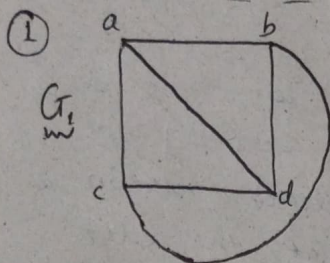
- f is one-one and onto function.
- for all $a, b \in V_1$, $\{a, b\} \in E_1$ if and only if $\{f(a), f(b)\} \in E_2$

When such a function exists, G_1 and G_2 are called Isomorphic graphs.

The prerequisites for G_1 and G_2 to be isomorphic are:-

- No. of vertices should be same.
- No. of edges should be same.
- Degree sequence of G_1 and G_2 should be same.
- No. of cycles should be same.

Check whether the following graphs are isomorphic or not



* No. of vertices:-

$$G_1:- 4 \quad G_2:- 4 \quad \text{same}$$

* No. of edges:-

$$G_1:- 6 \quad G_2:- 6 \quad \text{same}$$

* Degree sequence

$$G_1:- \langle 3, 3, 3, 3 \rangle \quad G_2:- \langle 3, 3, 3, 3 \rangle$$

same

* No. of cycles is same

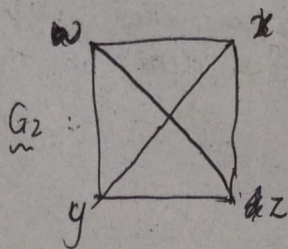
∴ Mapping

$$f(a) = w$$

$$f(b) = x$$

$$f(c) = y$$

$$f(d) = z$$



Edges:-

$$\{a, b\} = \{f(a), f(b)\} = \{w, x\}$$

$$\{a, c\} = \{f(a), f(c)\} = \{w, y\}$$

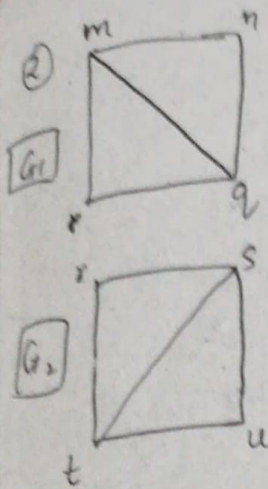
$$\{a, d\} = \{f(a), f(d)\} = \{w, z\}$$

$$\{b, c\} = \{f(b), f(c)\} = \{x, y\}$$

$$\{b, d\} = \{f(b), f(d)\} = \{x, z\}$$

$$\{c, d\} = \{f(c), f(d)\} = \{y, z\}$$

∴ G_1 and G_2 are isomorphic



Edges:

$$\{m, n\} = \{f(m), f(n)\} = \{s, r\}$$

$$\{m, p\} = \{f(m), f(p)\} = \{s, t\}$$

$$\{m, q\} = \{f(m), f(q)\} = \{s, t\}$$

$$\{n, q\} = \{f(n), f(q)\} = \{r, t\}$$

$$\{p, q\} = \{f(p), f(q)\} = \{u, t\}$$

* No. of vertices:

$$G_1: 4 \quad G_2: 4 \quad \text{Same}$$

* No. of edges:

$$G_1: 5 \quad G_2: 5 \quad \text{Same}$$

* Degree sequence:

$$G_1: \langle 2, 2, 3, 3 \rangle \quad G_2: \langle 2, 2, 3, 3 \rangle$$

Same

* No. of cycles is same

$$\therefore \text{Mapping: } f(m) = s$$

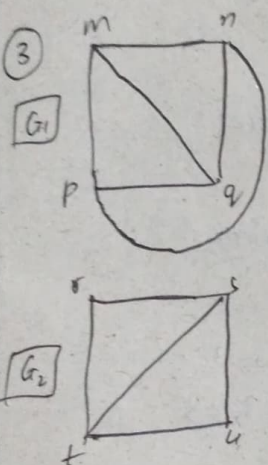
$$f(n) = r$$

$$f(p) = u$$

$$f(q) = t$$

$\therefore G_1$ and G_2 are

Isomorphic



No. of vertices:

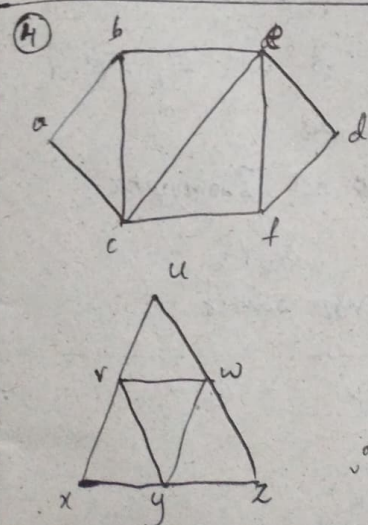
$$G_1: 4 \quad G_2: 4 \quad \text{Same}$$

No. of edges:

$$G_1: 6 \quad G_2: 5 \quad \text{Not same}$$

$\therefore G_1$ and G_2 are not Isomorphic

because no. of edges in G_1 & G_2 are not equal.



* No. of vertices

$$G_1: 6 \quad G_2: 6 \quad \text{Same}$$

* No. of edges

$$G_1: 9 \quad G_2: 9 \quad \text{Same}$$

* Degree sequence

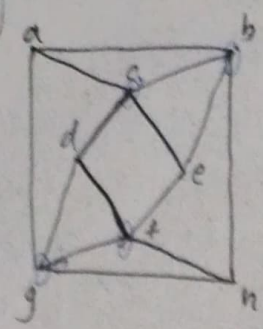
$$G_1: \langle 2, 2, 3, 3, 4, 4 \rangle$$

$$G_2: \langle 2, 2, 2, 2, 4, 4 \rangle \quad \text{Not same}$$

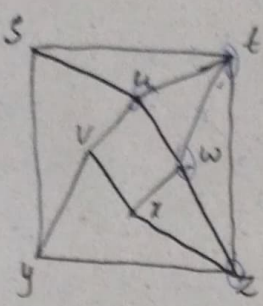
$\therefore G_1$ and G_2 are not Isomorphic

because Degree sequence of G_1 & G_2 are not equal.

5



* No. of vertices
 $G_1: 8$ $G_2: 8$ same
 * No. of edges
 $G_1: 14$ $G_2: 14$ same



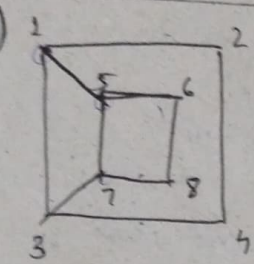
* Degree sequence
 $G_1: \langle 3, 3, 3, 3, 4, 4, 4, 4 \rangle$
 $G_2: \langle 3, 3, 3, 3, 4, 4, 4, 4 \rangle$

~~Both G_1 and G_2 are isomorphic~~
~~They are not isomorphic~~

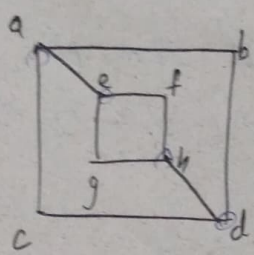
~~Both G_1 and G_2 have a cycle of length 4 exists~~

on vertices $t \rightarrow u \rightarrow w \rightarrow z \rightarrow t$ of degree 4 in G_2
 Such a cycle doesn't exist in G_1 so
 G_1 and G_2 are not isomorphic

6



* No. of vertices
 $G_1: 8$ $G_2: 8$ same
 * No. of edges
 $G_1: 10$ $G_2: 10$ same



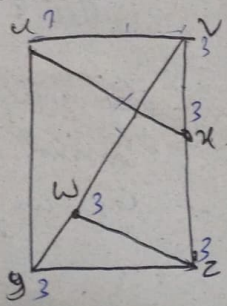
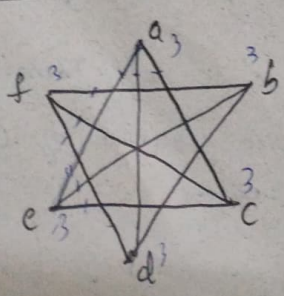
* Degree sequence
 $G_1: \langle 2, 2, 2, 2, 3, 3, 3, 3 \rangle$
 $G_2: \langle 2, 2, 2, 2, 3, 3, 3, 3 \rangle$

A cycle of length 4 exists on vertices $4 \rightarrow 5 \rightarrow 7 \rightarrow 3 \rightarrow 4$ of degree 3 in G_1 . But such a cycle doesn't exist in G_2 so

G_1 and G_2 are ~~same~~ not Isomorphic

No. of cycles of length 4 :- $G_1 = 3$ $G_2 = 2$
 $\therefore$ Not same.

7



* No. of vertices :- $G_1 :- 6$ $G_2 :- 6$ same

* No. of edges :- $G_1 :- 9$ $G_2 :- 9$ same

* Degree sequence :- $G_1 :- \langle 3, 3, 3, 3, 3, 3 \rangle$

$G_2 :- \langle 3, 3, 3, 3, 3, 3 \rangle$

* No. of cycles are same :-

~~isom~~ Mapping:

$f(a) = u$ $f(b) = z$ $f(c) = v$ $f(d) = y$ $f(e) = x$ $f(f) = w$

&

Edges:-

$$\{a, c\} = \{f(a), f(c)\} = \{u, v\}$$

$$\{a, d\} = \{f(a), f(d)\} = \{u, y\}$$

$$\{a, e\} = \{f(a), f(e)\} = \{u, x\}$$

$$\{b, d\} = \{f(b), f(d)\} = \{z, y\}$$

$$\{b, e\} = \{f(b), f(e)\} = \{z, x\}$$

$$\{b, f\} = \{f(b), f(f)\} = \{z, w\}$$

$$\{c, e\} = \{f(c), f(e)\} = \{v, x\}$$

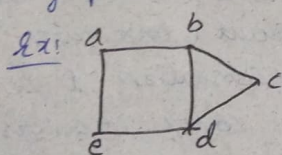
$$\{c, f\} = \{f(c), f(f)\} = \{v, w\}$$

$$\{d, f\} = \{f(d), f(f)\} = \{y, w\}$$

Theorems:

* If G is undirected or multi-graph then $\sum_{v \in V} \deg(v) = 2|E|$

Proof: as we consider the edge $\{a, b\} \in G$ we find that the edge contributes a count of 1 to each of the vertices i.e. to each of $\deg(a)$ and $\deg(b)$ consequently a count of 2 to $\sum_{v \in V} \deg(v)$ is to the degree of a particular graph.



$$\text{degree} = 2+2+2+3+2 = 12$$

$$\text{edges} = 6$$

$$\therefore \text{degree} = 2 \times \text{edges}$$

Ex: How many edges are there in a graph with 10 vertices & each of the degree of vertices is 6.

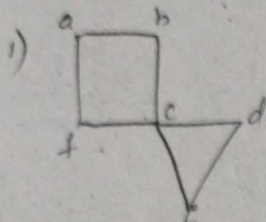
so $\therefore \text{degree} = 6 \times 10 = 60$ $\therefore \text{edges} = \frac{\text{degree}}{2} = 30$

Euler's Circuit

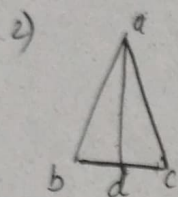
Euler's circuit is a circuit where all ~~the~~ edges are visited once.

If G has a Euler's circuit then it is 'Eulerian Graph'.

If G has a Euler's trail then it is called 'Semi-Eulerian Graph'.



$a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow c \rightarrow f \rightarrow a$



Not a Eulerian Graph but a semi Euler Graph

$a \rightarrow c \rightarrow d \rightarrow b \rightarrow a \rightarrow d$

* Let G be the undirected graph with no isolated vertex then G has Euler's circuit iff G is connected and every vertex in G has even degree.

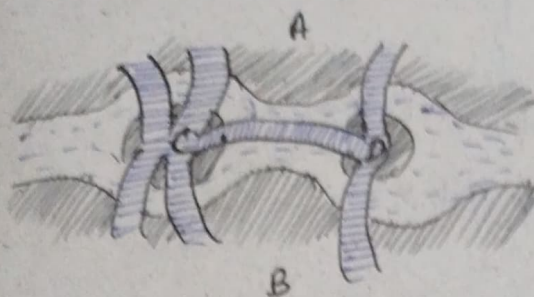
Proof:-

Let us say G has an Euler circuit then the circuit begins with vertex ~~and~~ 'a' and continues with an edge incident to 'a' say $\{a, b\}$. This edge contributes 1 to degree of 'a' and 1 to degree of 'b'. Each edge in the circuit when passes through the vertex it contributes 2 to each vertex degree. Therefore circuit enters through the edge incident with that edge and leaves through another such edge and finally the circuit terminates where it has started again contribution 1 to degree of 'a'. Therefore each vertex in Euler's circuit has even degree.

* If G is undirected graph with no ~~isolated~~ isolated vertices then we can construct a Euler's trail if and only if G is connected & has exactly 2 vertices of odd degree.

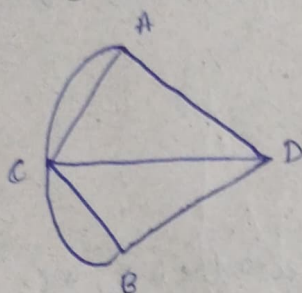
KONIGSBERG

BRIDGE PROBLEM



There are 4 land areas which are connected to each other by 7 bridges. The problem was to start at any of the 4 land areas of the city and cross over each of the 7 bridges

exactly once and return to the starting point.



Euler represented the bridge problem as a graph as shown.

The vertices represent the land areas and the edges represent the bridge. Then he found that there are 3 vertices with degree 3 and one vertex with degree 5. We

know that in order to have a Euler's circuit all the vertices must have an even degree. Hence there is no circuit in the graph.

Algorithm to ~~find~~ Construct an Euler's circuit

1) Circuit :-

A circuit in graph G beginning with any arbitrary vertex with the edges added to form a ~~path~~ path that returns to the same ~~start~~ vertex.

~~Let H be the~~

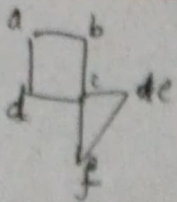
2) H :- G with the edges of the circuit removed.

3) Subcircuit :- A circuit at H beginning at a arbitrary vertex in H with the edges added to form a ~~path~~ path E_1 return to the vertex.

4) H :- with edges of the subcircuit removed.

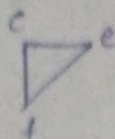
5) Circuit :- circuit with subcircuit inserted at appropriate vertex.

Ex1.



circuit: $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$

H1:

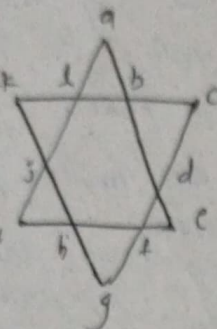


subcircuit:

$c \rightarrow e \rightarrow f \rightarrow c$

$a \rightarrow b \rightarrow c \rightarrow e \rightarrow f \rightarrow c \rightarrow d \rightarrow a$
subcircuit

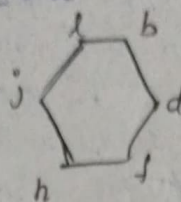
Ex1.



circuit:

$a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow g \rightarrow h \rightarrow i \rightarrow j \rightarrow k \rightarrow l \rightarrow a$
 $\rightarrow l \rightarrow a$

H1:

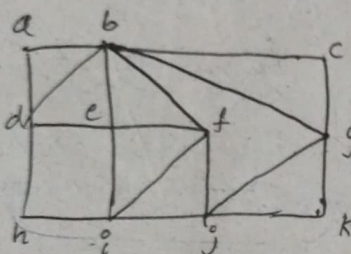


Subcircuit:

$l \rightarrow b \rightarrow d \rightarrow f \rightarrow h \rightarrow j \rightarrow l$

$\therefore a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow g \rightarrow h \rightarrow i \rightarrow j \rightarrow k \rightarrow l \rightarrow b \rightarrow d \rightarrow f \rightarrow h \rightarrow j \rightarrow l \rightarrow a$

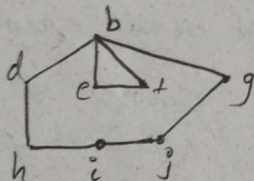
Ex1.



circuit:

$a \rightarrow b \rightarrow c \rightarrow g \rightarrow k \rightarrow j \rightarrow f \rightarrow i \rightarrow e \rightarrow d \rightarrow a$

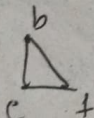
H1:



subcircuit:

$b \rightarrow d \rightarrow h \rightarrow i \rightarrow j \rightarrow g \rightarrow b$

H1:



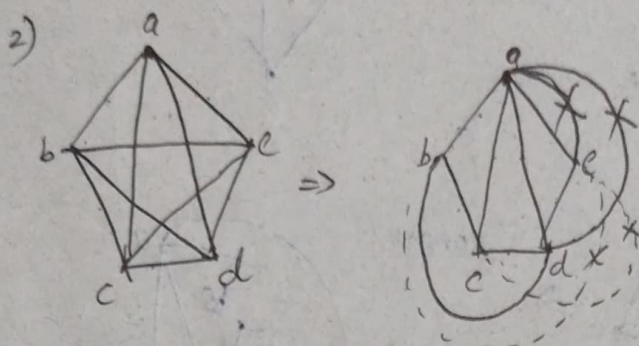
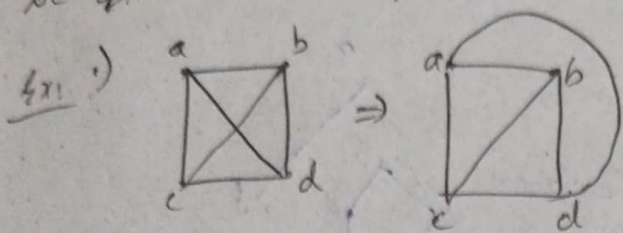
subcircuit:

$b \rightarrow c \rightarrow f \rightarrow b$

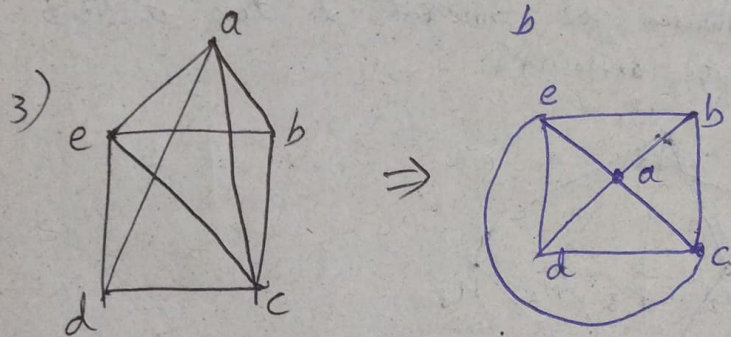
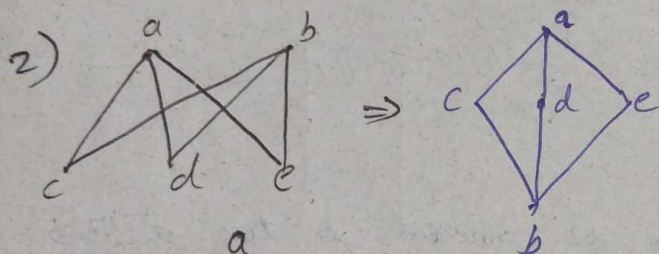
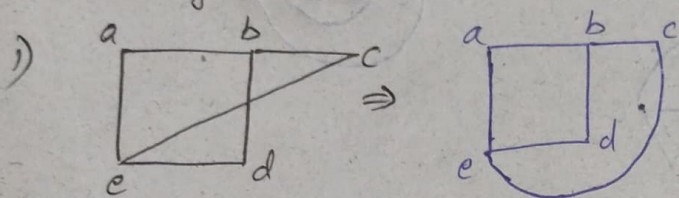
$\therefore a \rightarrow b \rightarrow e \rightarrow f \rightarrow b \rightarrow d \rightarrow h \rightarrow i \rightarrow j \rightarrow g \rightarrow b \rightarrow c \rightarrow g \rightarrow k \rightarrow j \rightarrow f \rightarrow i \rightarrow e \rightarrow d \rightarrow a$

Planar Graph

A graph is said to be planar graph if it can be drawn on a plane without edge crossing.

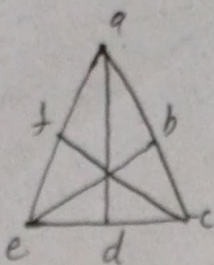


Ex: Draw the given planar graphs without edge crossing.



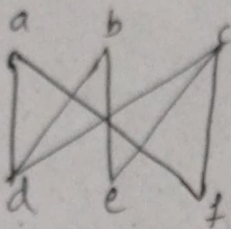
* Determine if the given graphs are planar or not. If so draw them without edge crossing.

1)

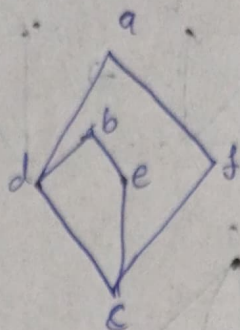


Not planar.

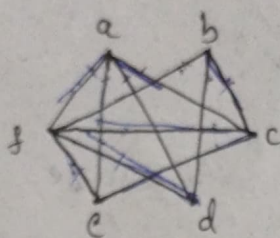
2)



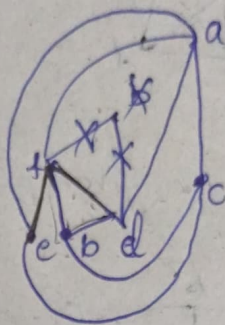
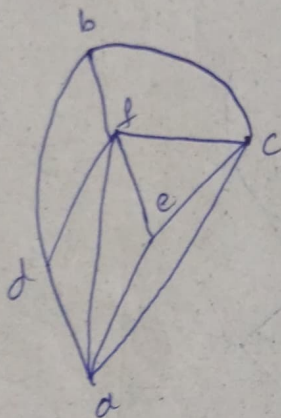
planar!



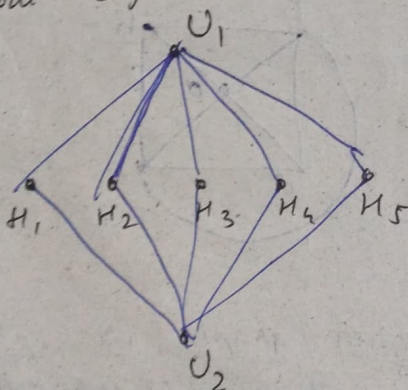
3)



planar!



4) Can ~~the~~ 5 houses be connected to two utilities without edge crossing



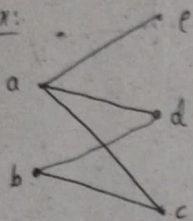
Bipartite Graph

A Graph in which $V = V_1 \cup V_2$ and $V_1 \cap V_2 = \emptyset$ and every edge of graph G of the form $\{a, b\}$ with $a \in V_1$ and $b \in V_2$

* Complete Bipartite Graph

If each vertex of V_1 is joined with every vertex of V_2 then we have complete bipartite graph $K_{m,n}$ where m is the no. of vertices in vertex set V_1 and n is the no. of vertices in vertex set V_2 .

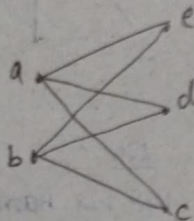
Ex:



Bi-partite graph

$$V_1 = \{a, b\}$$

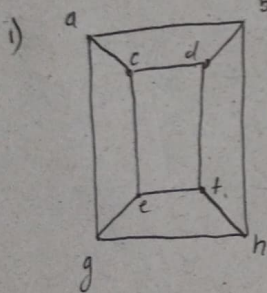
$$V_2 = \{c, d, e\}$$



complete bi-partite graph

$$K_{2,3}$$

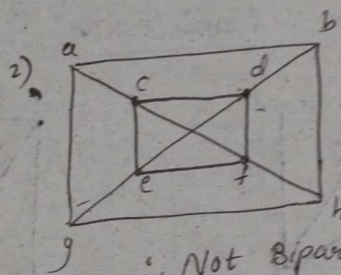
Determine whether the following graphs are bipartite or not



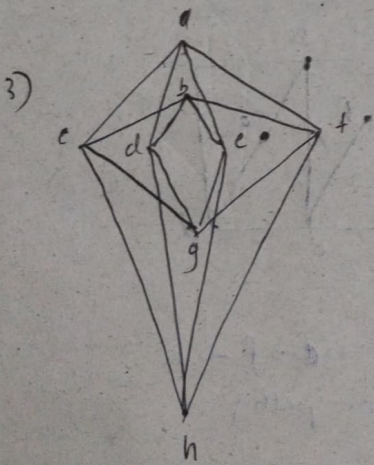
$$V_1 = \{a, c, d, h\}$$

$$V_2 = \{b, e, f, g\}$$

$\therefore$ Bipartite Graph



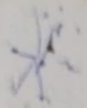
$\therefore$ Not Bipartite



$$V_1 = \{a, b, g, h\}$$

$$V_2 = \{c, d, e, f\}$$

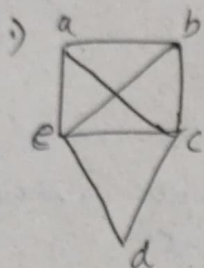
$\therefore$ Complete Bipartite ($K_{4,4}$)



Hamilton path and cycle

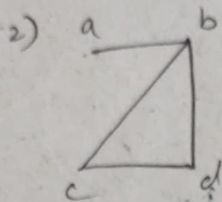
If G is a graph or a multigraph & cardinality of vertex of G is ≥ 3 , we say that G is hamiltonian graph if there is a cycle in G which contains every vertex in graph G .

Determine which of the following has hamiltonian cycle or path



Yes:

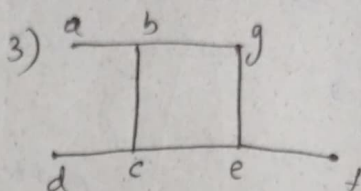
$a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow a$



No cycle

But has hamiltonian path

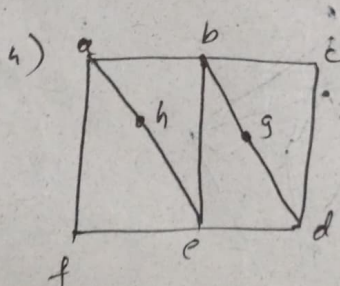
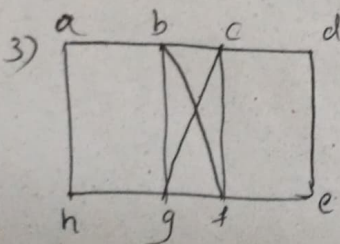
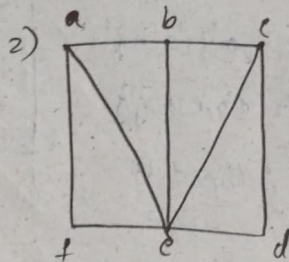
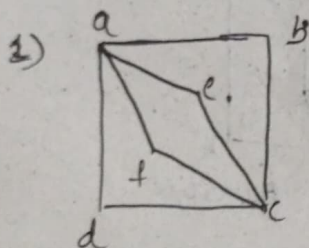
$a \rightarrow b \rightarrow c \rightarrow d$



No

No path or cycle

Determine which of the following graphs are eulerian or hamiltonian :-



1) Eulerian :- $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a \rightarrow e \rightarrow f \rightarrow a$
Not hamiltonian (no cycle & no path)

2) ~~Eulerian~~ Not eulerian (no circuit & no trail)
hamiltonian $a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow g \rightarrow h \rightarrow a$

3) Eulerian

$a \rightarrow b \rightarrow f \rightarrow c \rightarrow g \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow g \rightarrow h \rightarrow a$

Hamiltonian

$a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow g \rightarrow h \rightarrow a$

4) Euler's trail (Not ~~egg~~ circuit)

~~$a \rightarrow b \rightarrow c \rightarrow d \rightarrow g \rightarrow b \rightarrow e \rightarrow h \rightarrow a \rightarrow f \rightarrow e \rightarrow d$~~
semi eulerian

Hamiltonian path (Not cycle)

$f \rightarrow a \rightarrow h \rightarrow e \rightarrow b \rightarrow g \rightarrow d \rightarrow c$

semi hamiltonian

GRAPH COLORING

Proper coloring of a graph :-

It is such that no two adjacent vertices have same color.

Chromatic number

$\chi(G)$:- It is the minimum no. of colors required to properly color the graph is nothing but the chromatic number of G .

* A Graph G is said to be n -colorable if we can properly colour it k no. of colours. The graph is said to be k -chromatic.

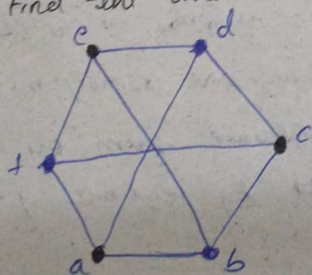
Notes 1) A Graph consisting of only isolated vertices is 1-chromatic.

2) A Graph with 1 or more edges is atleast 2-chromatic

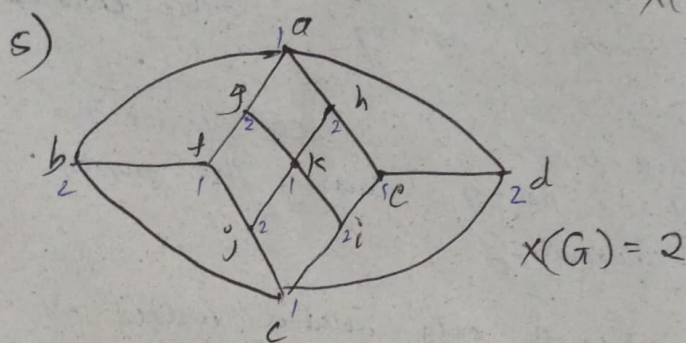
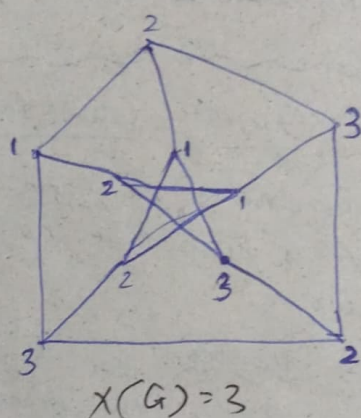
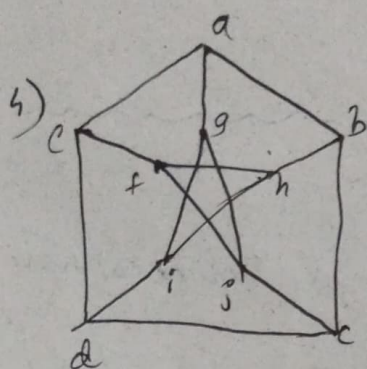
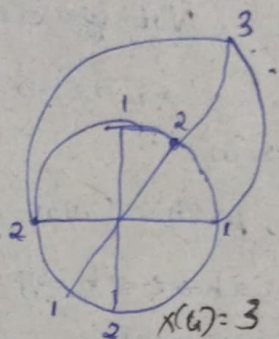
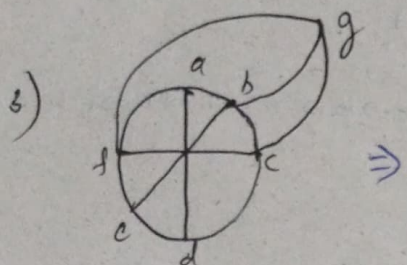
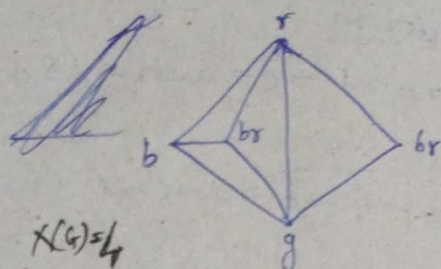
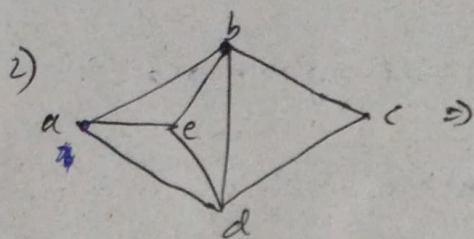
3) If G is a graph with n vertices then

4) The Chromatic number of complete graph is n .
ie $\chi(K_n) = n$

* Find the chromatic number of following graphs



$$\chi(G) = 2$$



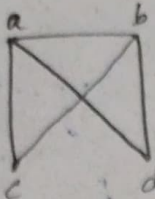
Homeomorphic Graphs

Let G be the loop free undirected graph where $|E| \neq 0$. An elementary subdivision of G results when an edge $\{u, v\}$ is removed from the graph and the edges $\{u, w\}$ & $\{w, v\}$ are added to $G - e$. The loop free undirected graphs G_1 & G_2 are called homeomorphic if they are isomorphic or if both of them can be obtained from by same loop free undirected graph H by sequence of elementary subdivisions.

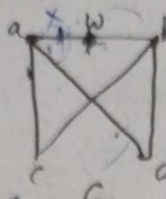
Example:

R

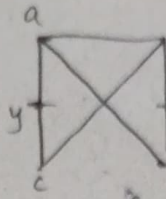
9



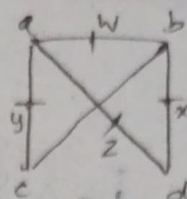
G_1



G_2



G_3



G_4

1 subdivision

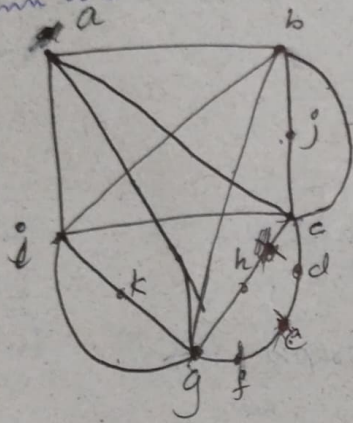
2 subdivision

4 subdivision

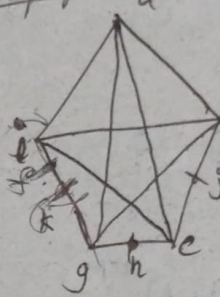
Kuratowski's theorem

A graph is non planar if it contains a subgraph i.e. ~~home~~ homeomorphic to either K_5 or $K_{3,3}$

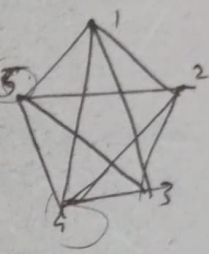
Example:-



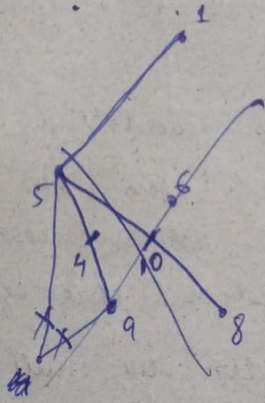
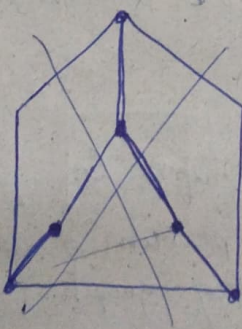
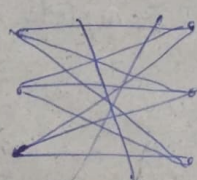
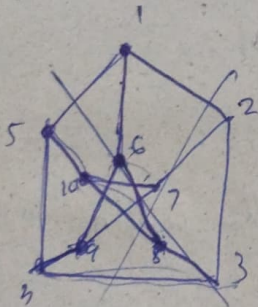
subgraph :- a



K_5

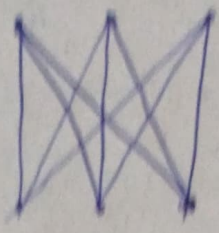
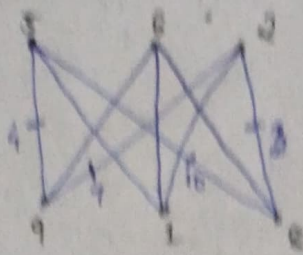


Check whether peterson graph is planar or not



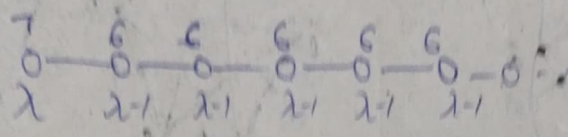
5, 6, 2

9, 1, 8

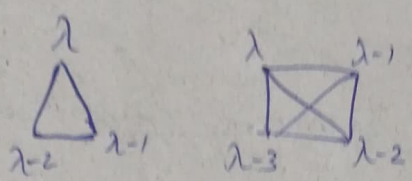


$\therefore$ 4 elementary subdivisions on $K_{2,3}$ lead to
 a subgraph of Petersen graph
 $\therefore$ Petersen graph is not planar

Chromatic Polynomial



$\therefore \lambda(\lambda-1)^{n-1}$ Path Graph



λC_n λK_n λN_n Complete graph

λ^n Isolated graph

Given a connected graph G and λ number of different colors, the problem of find no. of different ways of properly coloring graph G with λ colors is called as chromatic polynomial

1) Consider the null Graph (N_n) with n vertices. In this graph no 2 vertices are adjacent $\therefore$ A proper coloring of this graph can be done by

assigning a single colour to all the vertices. If λ no. of colours are available, each vertex of the graph has λ possible choices of colours assigned to it. As such graph can be properly coloured with λ^n different ways.

2) Consider the complete Graph K_n . In this case, every two vertices are adjacent. As such there must be atleast n colours for proper colouring of K_n . If no. of different colours available is λ then the no. of

the 0 if $\lambda < n$

1 if $\lambda = n$

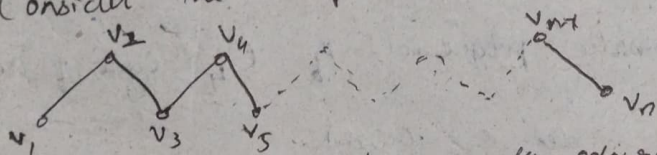
> 1 if $\lambda > n$

Let $v_1, v_2, v_3, \dots, v_n$ be the vertices and $\lambda > n$ for proper coloring of K_n . The vertex v_1 can be assigned any of the λ colours. The vertex v_2 can be assigned any of the remaining $\lambda - 1$ colours

$$\begin{aligned} \therefore v_1 &= \lambda \\ v_2 &= \lambda - 1 \\ &\vdots \\ v_n &= \lambda - n + 1 \end{aligned} \quad \therefore \lambda(\lambda-1)(\lambda-2)\dots(\lambda-n+1)$$

$$\text{or } (\lambda - n)! \binom{\lambda}{n}$$

3) Consider the Graph L_n (Path Graph) of n vertices



This graph cannot be properly coloured with 1 colour but can be properly coloured with 2. By assigning 1 colour to all odd vertices & other colour to all even vertices

If $\lambda > 2$ the proper colouring of this graph.

The vertex v_1 can be assigned any one of the λ colours & each of the remaining vertices can be assigned only any one of the $\lambda - 1$ colors. So the polynomial is

$$\lambda(\lambda-1)^{n-1}$$

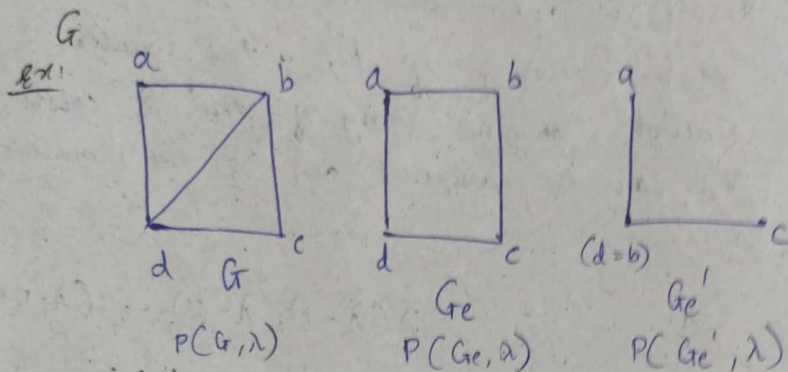
$$\therefore C(K_n, \lambda) = \lambda(\lambda-1)(\lambda-2) \dots (\lambda-n+1) \quad \lambda \geq n$$

$$C(L_n, \lambda) = \lambda(\lambda-1)^{n-1} \quad \lambda \geq 2$$

$$C(N_n, \lambda) = \lambda^n$$

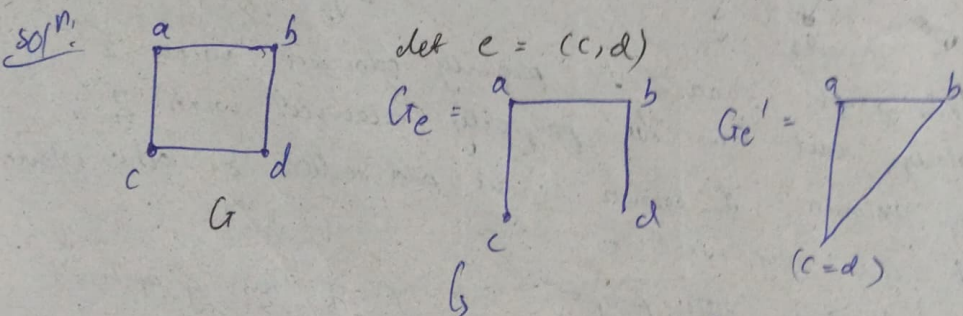
Decomposition Theorem

Let G be the graph and e be the edge of the graph, $G_e = G - e$ be the subgraph of G which is obtained by deletion of edge e from G without deleting the vertices and we construct a new graph G_e' by identifying & merging the vertices a and b in G_e . Then G_e' is a subgraph of G_e as well as



G is a connected graph and e is the edge of G , then $P(G, \lambda) = P(G_e, \lambda) - P(G_e', \lambda)$

* Find the chromatic polynomial of C_4 (cycle of length 4)



$$P(G_e, \lambda) = \lambda(\lambda-1)^3$$

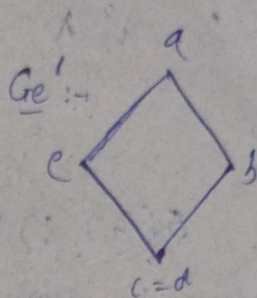
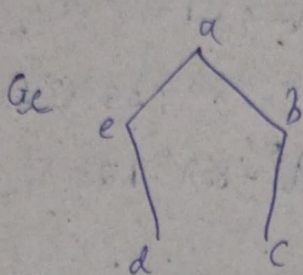
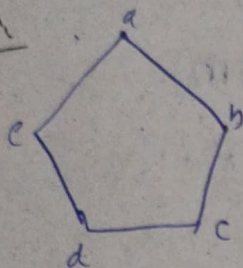
$$P(G_e', \lambda) = \lambda(\lambda-1)(\lambda-2)$$

$$\therefore P(G, \lambda) = P(G_e, \lambda) - P(G_e', \lambda)$$

$$= \lambda(\lambda-1)^3 - \lambda(\lambda-1)(\lambda-2)$$

$$\begin{aligned} & \lambda(\lambda-1)(\lambda^2-2\lambda+1-\lambda+2) \\ P(G, \lambda) &= \lambda(\lambda-1)(\lambda^2-3\lambda+3) \\ &= \lambda(\lambda^3-4\lambda^2+6\lambda-3) \\ &= \lambda^4-4\lambda^3+6\lambda^2-3\lambda \end{aligned}$$

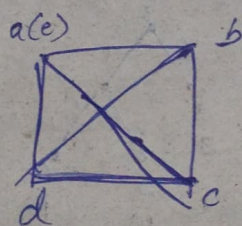
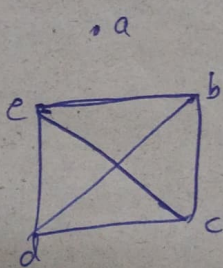
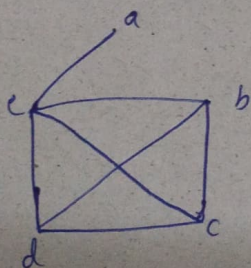
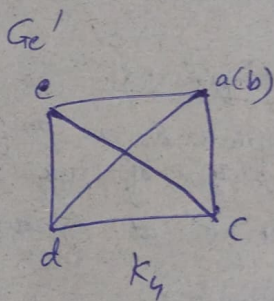
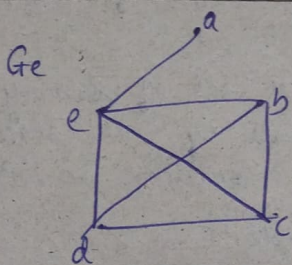
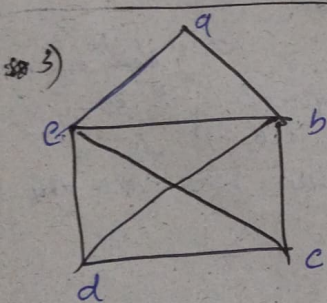
soln



Now $P(Ge', \lambda) = \lambda^4 - 4\lambda^3 + 6\lambda^2 - 3\lambda$

$$P(Ge, \lambda) = \lambda(\lambda-1)^4$$

$$\begin{aligned} \therefore P(G, \lambda) &= \lambda(\lambda-1)^4 - \lambda^4 + 4\lambda^3 - 6\lambda^2 + 3\lambda \\ &= \lambda(\lambda^2-2\lambda+1)^2 - \lambda^4 + 4\lambda^3 - 6\lambda^2 + 3\lambda \\ &= \lambda(\lambda^4 + 4\lambda^2 + 1 - 4\lambda^3 - 4\lambda + 2\lambda^2) - \lambda^4 + 4\lambda^3 - 6\lambda^2 + 3\lambda \\ &= \lambda^5 - 5\lambda^4 + 10\lambda^3 - 10\lambda^2 + 4\lambda \end{aligned}$$



$$\begin{aligned} &= \lambda^2(\lambda-1)(\lambda-2)(\lambda-3) - \lambda(\lambda-1)(\lambda-2)(\lambda-3) \\ P(Ge, \lambda) &= \lambda(\lambda-1)^2(\lambda-2)(\lambda-3) \end{aligned}$$

$$P(Ge', \lambda) = \lambda(\lambda-1)(\lambda-2)(\lambda-3)$$

$$\therefore P(Ge, \lambda) = \lambda(\lambda-1)^2(\lambda-2)(\lambda-3) - \lambda(\lambda-1)(\lambda-2)(\lambda-3)$$

$$= \lambda(\lambda-1)(\lambda-2)^2(\lambda-3)$$

$$\therefore \text{Let } \lambda = \lambda^2 - 4\lambda + 3 \quad (\lambda^2 - 4\lambda + 4)$$

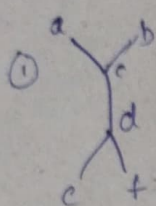
$$\lambda(\lambda^4 - 8\lambda^3 + 23\lambda^2 - 28\lambda + 12)$$

$$= \lambda^5 - 8\lambda^4 + 23\lambda^3 - 28\lambda^2 + 12\lambda$$

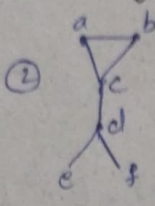
5(4)(3)(2)

For 5 colors 360 //

Trees

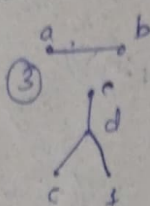


Tree

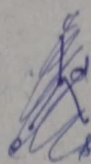


Graph

Not
tree.



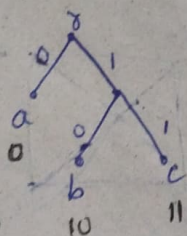
Forest.



Rooted trees: A Graph G is called as a rooted tree if there is a unique vertex ' r ' which is called as a root in G with in degree of ' r ' is zero (0)

Prefix code: It is a set of binary sequence. Let set P be a binary sequence. S is called as a prefix code if no sequence in P is a prefix of any other sequence in P . Then we call this set P as prefix code.

Example: $a:0 \quad b:10 \quad c:11 \quad P =$



Check whether the following are prefix code or not

1) a: 01 e: 0 n: 1101 y: 10 t: 1

ada
ada:- 01101 but $\frac{0}{e} \frac{1101}{n}$ Not a prefix code

2) a: 111 e: 0 n: 1100 y: 1101 t: 10
Yes a prefix code

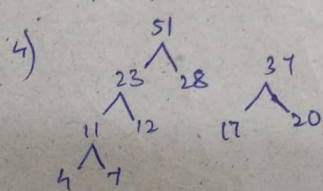
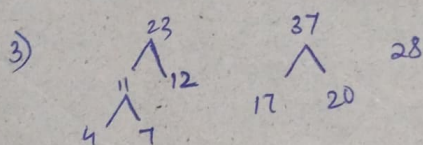
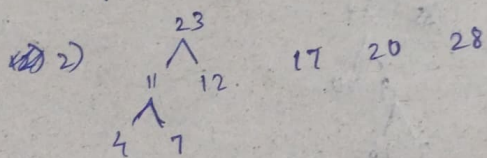
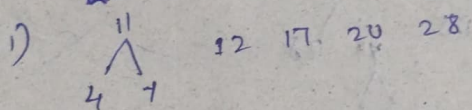
Huffman's coding algorithm

Step 1: Assign the given weights one each to set 'S' of m isolated vertices. while $|S| \geq 1$, then perform the following

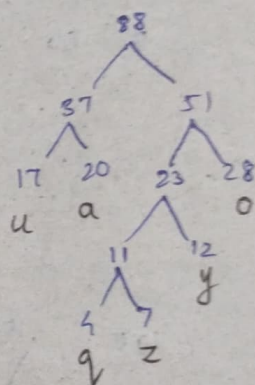
- (1) Find 2 trees T and T' in S with smallest 2 ~~best~~ weights w and w' respectively
- (2) Create a new complete binary tree T^* with root weight as $w^* = w + w'$ and having T and T' as left and right subtrees respectively.
- (3) Place T^* in S and delete T and T'

Example consider the optimal prefix code for the symbol a, o, q, u, y, z that occurs with the frequencies 20, 28, 4, 17, 12, 7

sol: 4 7 12 17 20 28



$\therefore T =$



a: 01
o: 11
q: 1000
u: 00
y: 101
z: 1001

a: 01
o: 11
q: 1000
u: 00
y: 101
z: 1001

2) Construct a optimal prefix code for the symbols a to j that occur with the frequency

~~78~~ 78, 16, 30, 35, 125, 31, 20, 50, 80, 3
a b c d e f g h i j

1) 3, 16, 20, 30, 31, 35, 50, 78, 80, 125

2)

```

      19
     / \
    3   16
  
```

20 30 31 35 50 78 80 125

3)

```

      39
     / \
    19  20
   / \
  3   16
 
```

30 31 35 50 78 80 125

4)

```

      39
     / \
    19  20
   / \
  3   16
 
```

35 50 61 78 80 125

```

      61
     / \
    30  31
 
```

5)

```

      61
     / \
    30  31
 
```

50 74 78 80 125

```

      74
     / \
    35  39
   / \
  19  20
 / \
3   16
 
```

6)

```

      74
     / \
    35  39
   / \
  19  20
 / \
3   16
 
```

78 80 111 125

```

      111
     / \
    50  61
   / \
  30  31
 
```

7)

80 111 125 152

```

      111
     / \
    50  61
   / \
  30  31
 
```

```

      152
     / \
    74  78
   / \
  35  39
   / \
  19  20
 / \
3   16
 
```

8)

125 152 191

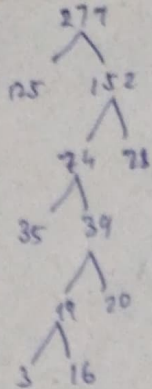
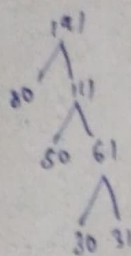
```

      152
     / \
    74  78
   / \
  35  39
   / \
  19  20
 / \
3   16
 
```

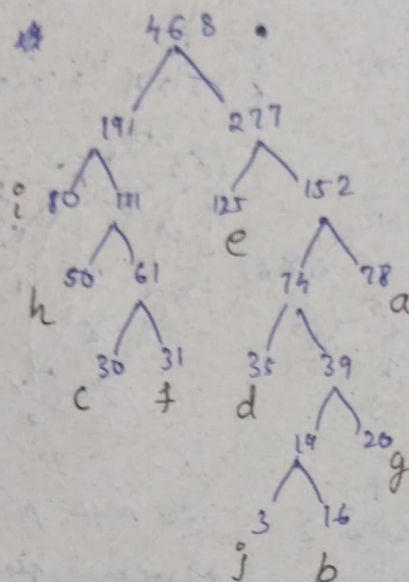
```

      191
     / \
    80  111
   / \
  50  61
   / \
  30  31
 
```

9)



T :-



a: 111

b: 110101

c: 0110

d: 1100

e: 10

f: 0111

g: 11011

h: 010

i: 00

j: 110100

Ford Fulkerson

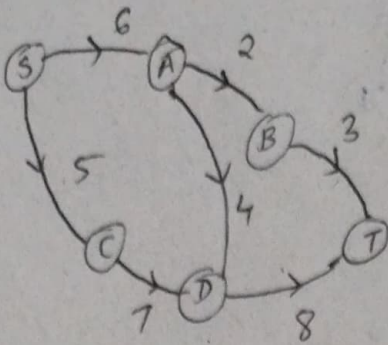
step 1:- If c is a path from source to sink then go to step-2 otherwise the optimal solution is determined by backward arcs.

step 2:- Calculate the maximum flow in the network

- * In the first step find the smallest capacity of the arc which is denoted by c .
- * In the second step subtract c from the capacity of each arc of the path
- * Third step:- Add backward edge for each arc in the path by adding c to the capacity of each backward arc in the path

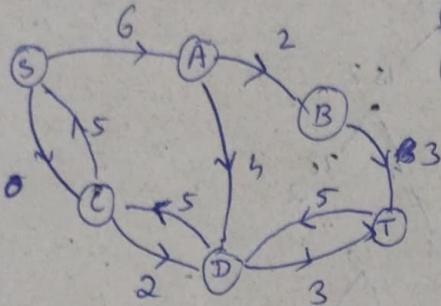
If there are still more paths go to step 1.

Ex 1

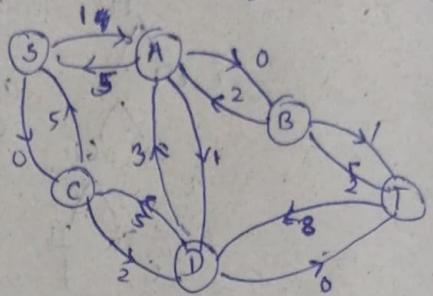


* ~~S → A → B → T~~ $S \rightarrow C \rightarrow D \rightarrow T$

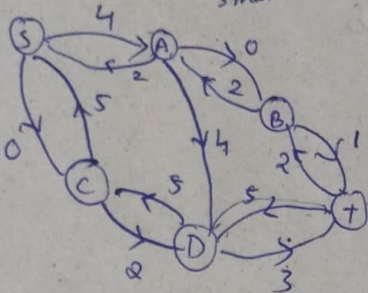
Smallest is 5 so



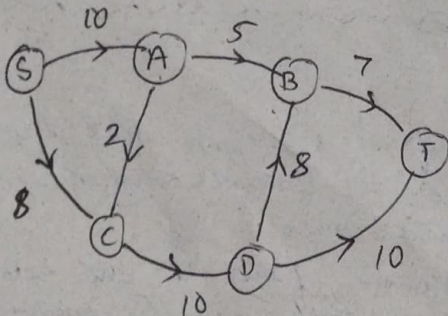
~~S → A → B → T~~ $S \rightarrow A \rightarrow D \rightarrow T$
smallest is 3



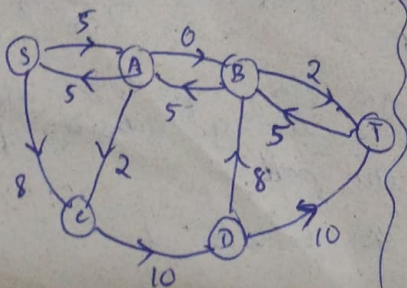
* $S \xrightarrow{6} A \xrightarrow{2} B \xrightarrow{3} T$
smallest is 2



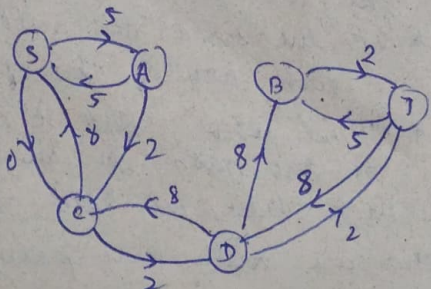
No more paths:
Maximum flow is 8
 $5 + 2 + 3 = 10$



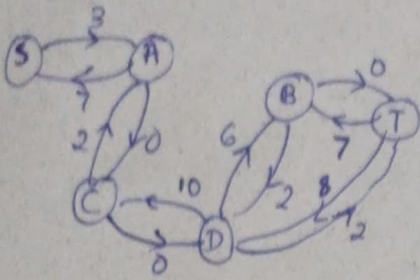
1) $S \xrightarrow{10} A \xrightarrow{5} B \xrightarrow{7} T$



2) $S \xrightarrow{8} C \xrightarrow{10} D \xrightarrow{10} T$

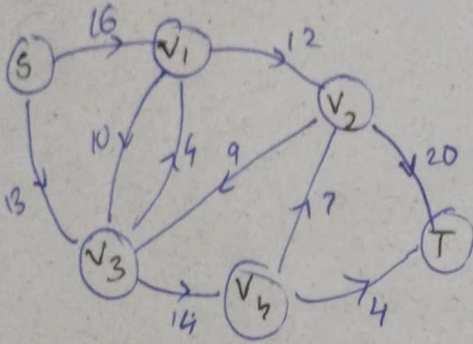


$$3) S \xrightarrow{5} A \xrightarrow{3} C \xrightarrow{2} D \xrightarrow{8} B \xrightarrow{2} T$$



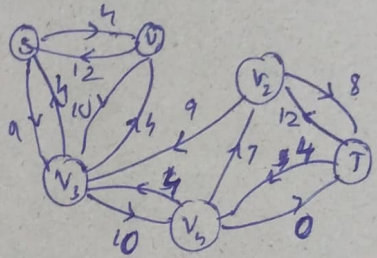
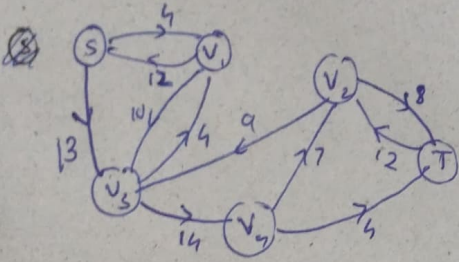
∴ No more paths

$$5 + 8 + 2 = 15 //$$

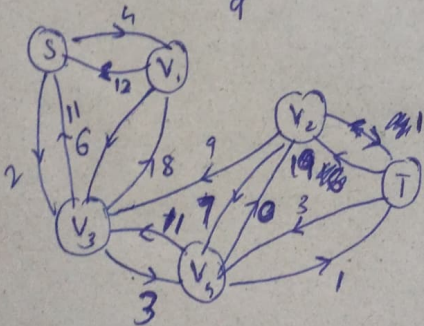


$$1) S \xrightarrow{16} V_1 \xrightarrow{10} V_3 \xrightarrow{9} V_2 \xrightarrow{20} T$$

$$2) S \xrightarrow{13} V_3 \xrightarrow{14} V_4 \xrightarrow{4} T$$



$$3) S \xrightarrow{11} V_3 \xrightarrow{10} V_4 \xrightarrow{7} V_2 \xrightarrow{8} T$$



No more paths

$$12 + 4 + 7 = 23 //$$

$$12 + 4 + 7 = 23 //$$